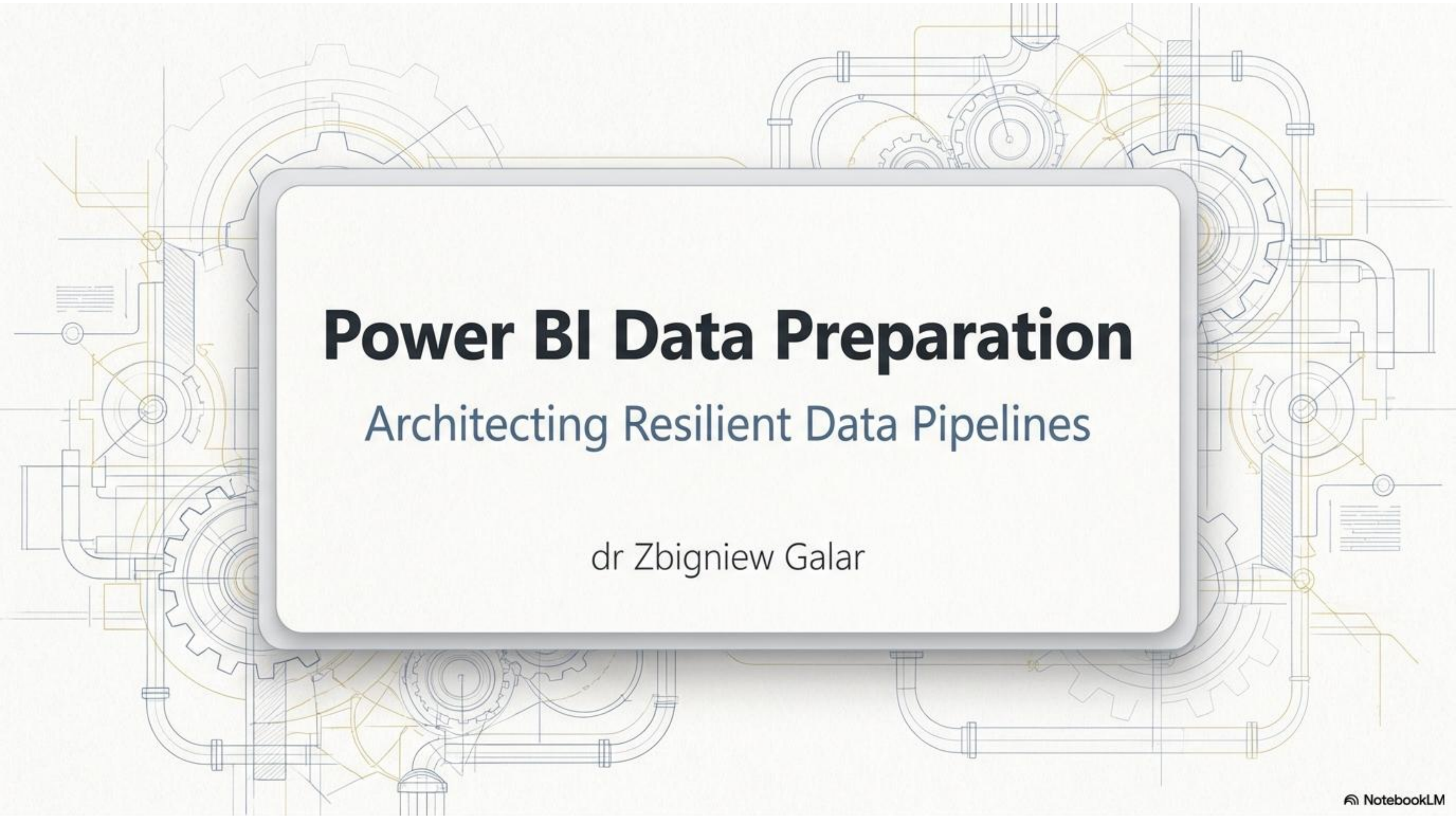


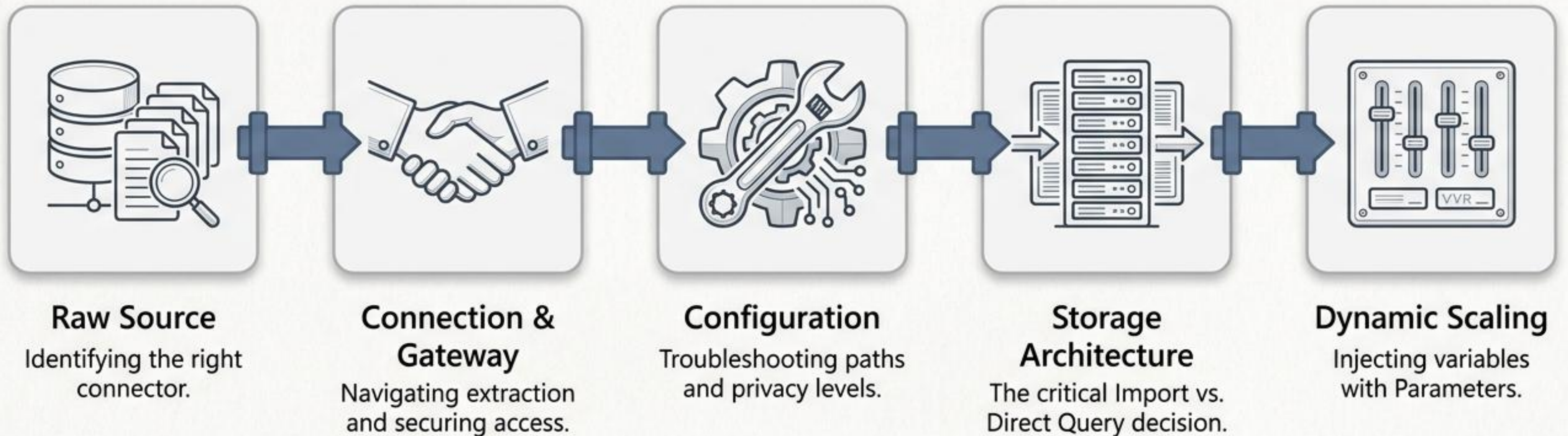
The background of the slide is a technical drawing or blueprint. It features several interlocking gears of different sizes, some with yellow highlights. A network of pipes and lines, also in yellow and grey, crisscrosses the background, creating a complex mechanical or industrial aesthetic. The overall color palette is light grey and white, with yellow accents.

Power BI Data Preparation

Architecting Resilient Data Pipelines

dr Zbigniew Galar

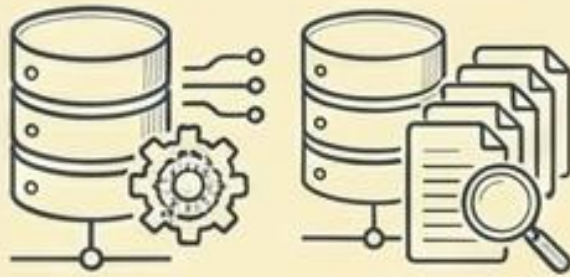
The Data Extraction Journey



Data preparation is not just a sequence of clicks; it is a series of strategic architectural choices.

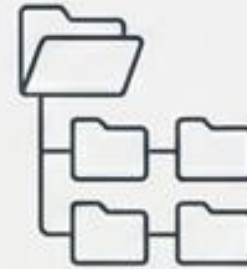
The Connector Ecosystem

Relational & Files



SQL Server (all flavors), Excel/CSV.

Bulk Extraction



Folders (SharePoint and File Explorer).

Microsoft Cloud



Power Platform (Dataverse, Dataflows) and Azure.

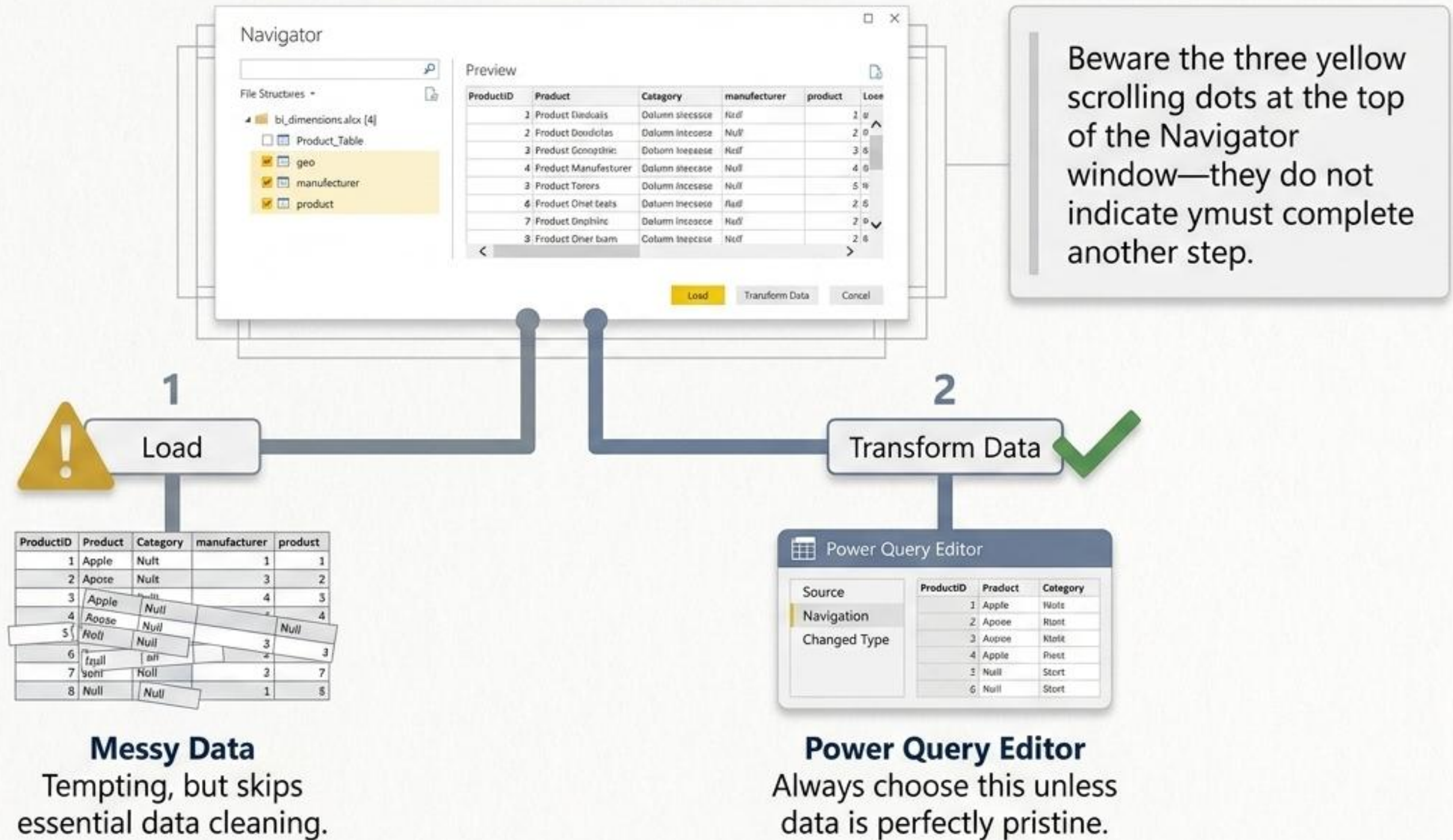
Live Models



Power BI datasets and Analysis Services.

Focus on the core: While Power BI offers hundreds of obscure connectors, mastering these primary categories is essential for everyday practice and the PL-300 exam.

The First Gateway: The Navigator Decision

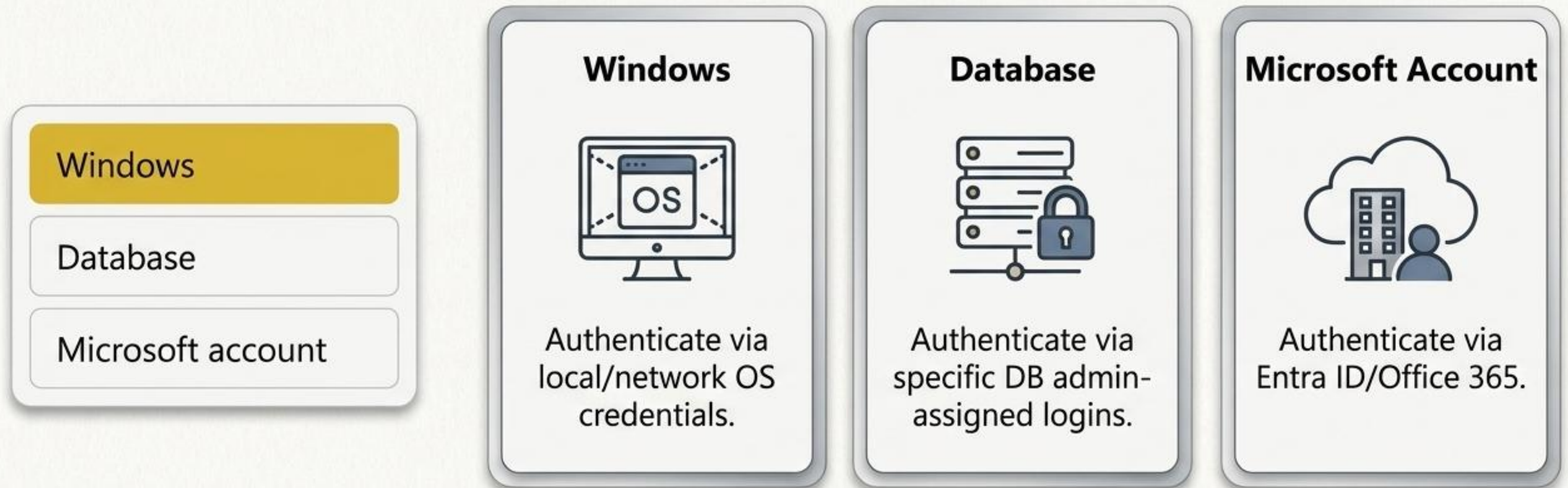


Connection Resilience: Fixing Broken Paths



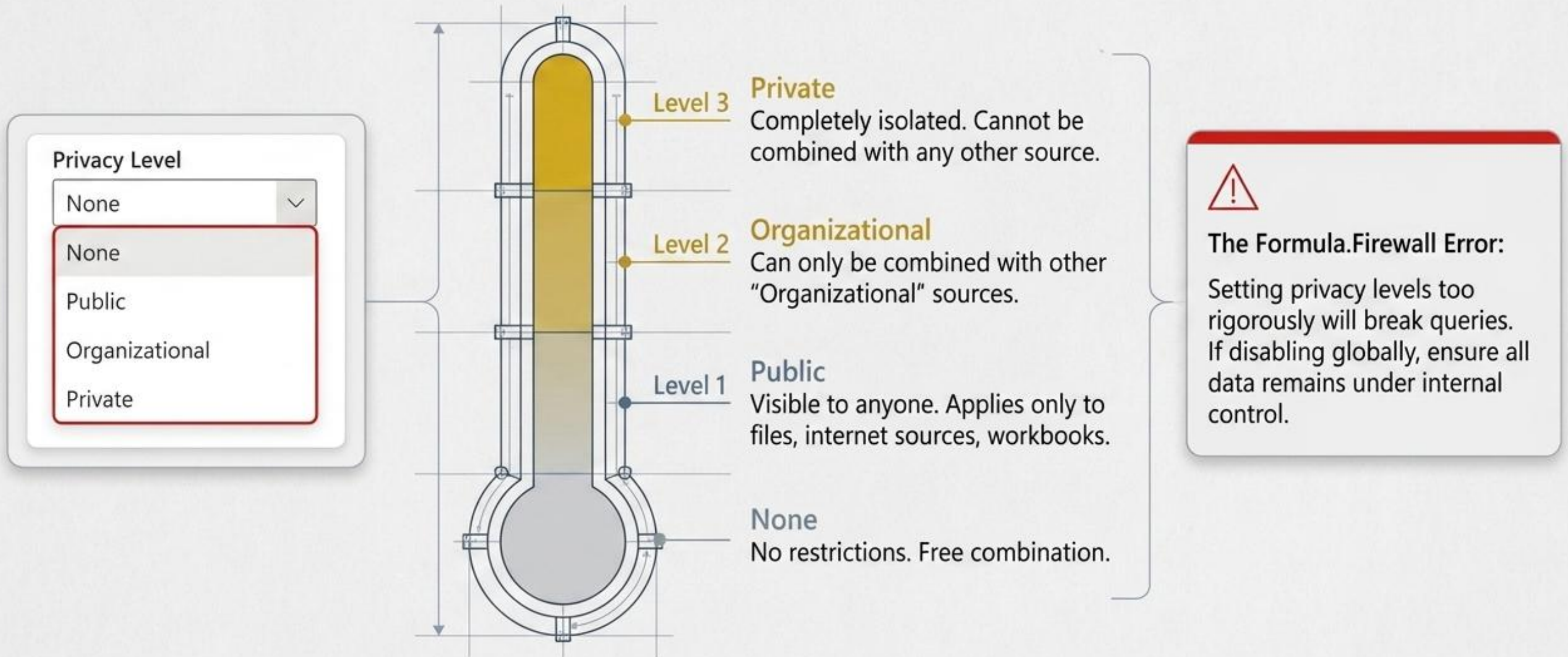
Key Insight: The Power Query Editor rigidly retains the metadata of the original file path. Re-mapping via Data Source Settings is the fastest path to resilience.

Authentication & Access Control

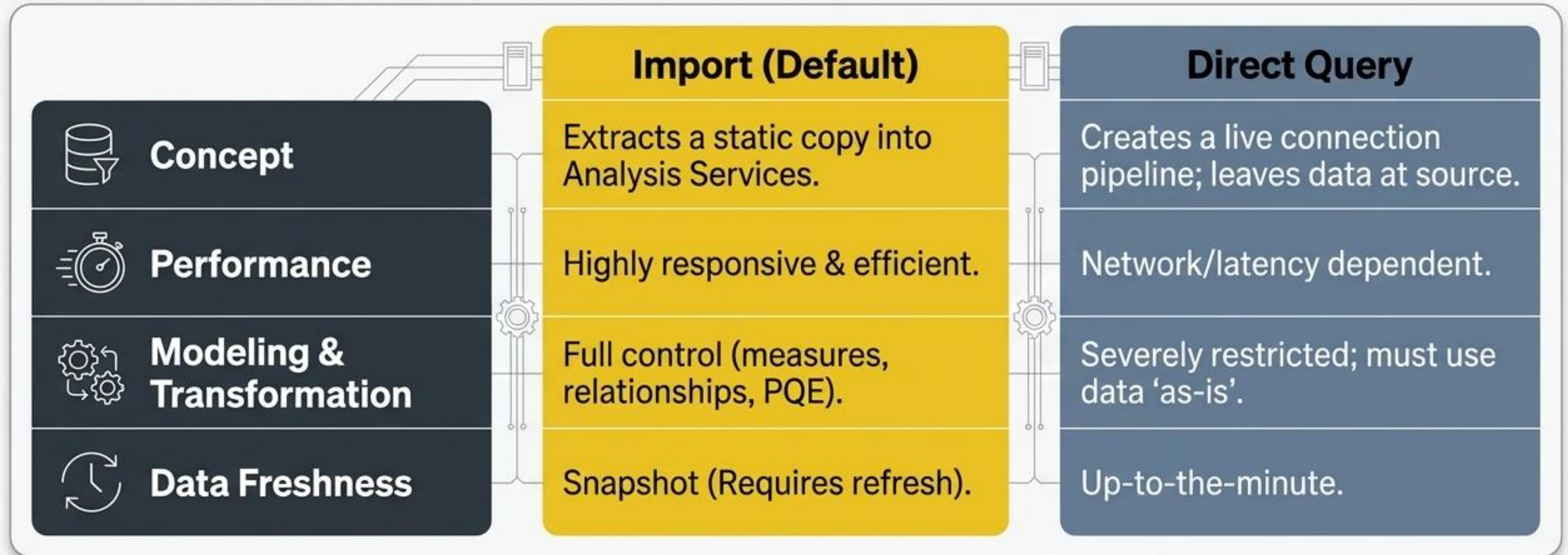


To update a changed password or switch accounts, navigate to:
Data Source Settings -> Edit Permissions -> Edit Credentials.

Data Privacy: The Leakage Risk Thermometer



The Storage Mode Showdown

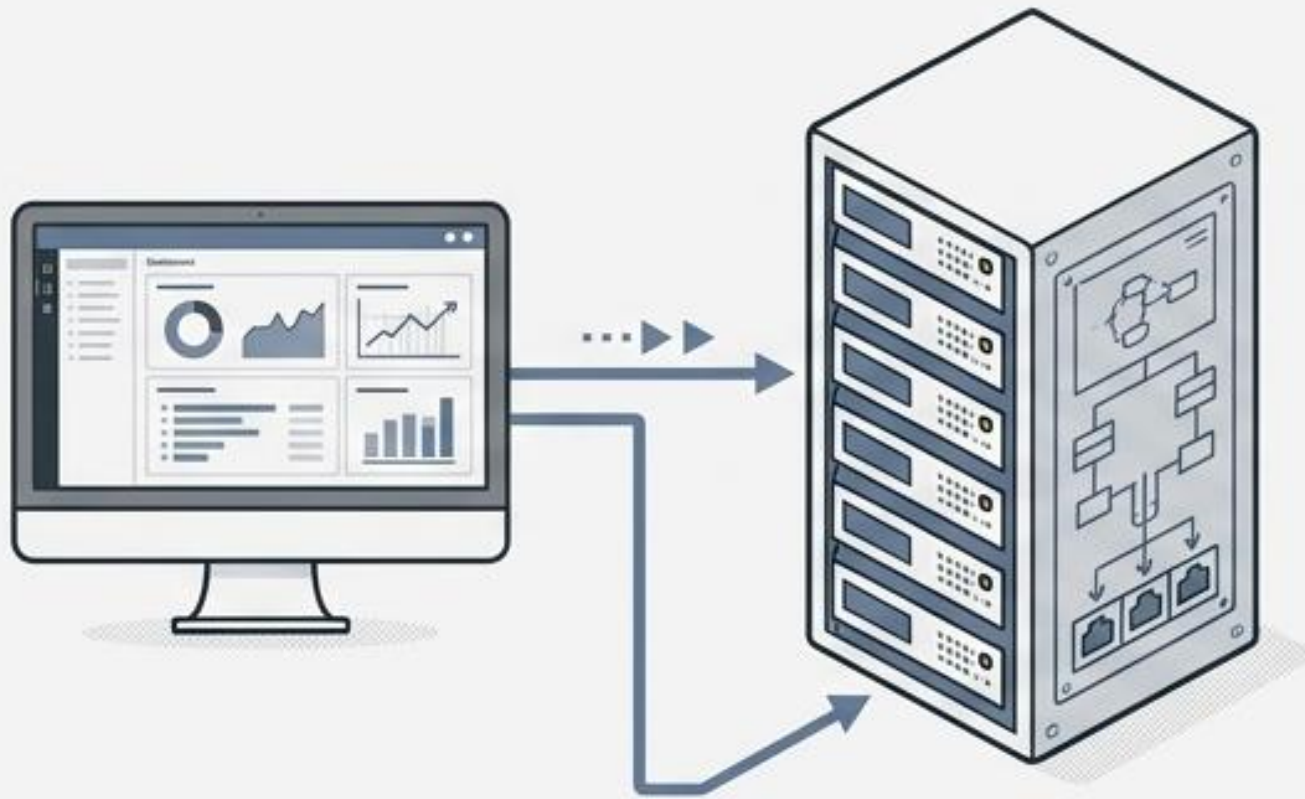


Diagnostic Checklist

Do you really need Direct Query? Ask: Are decisions made in real-time? Is historical data utilized? Do different sources refresh at different rates?

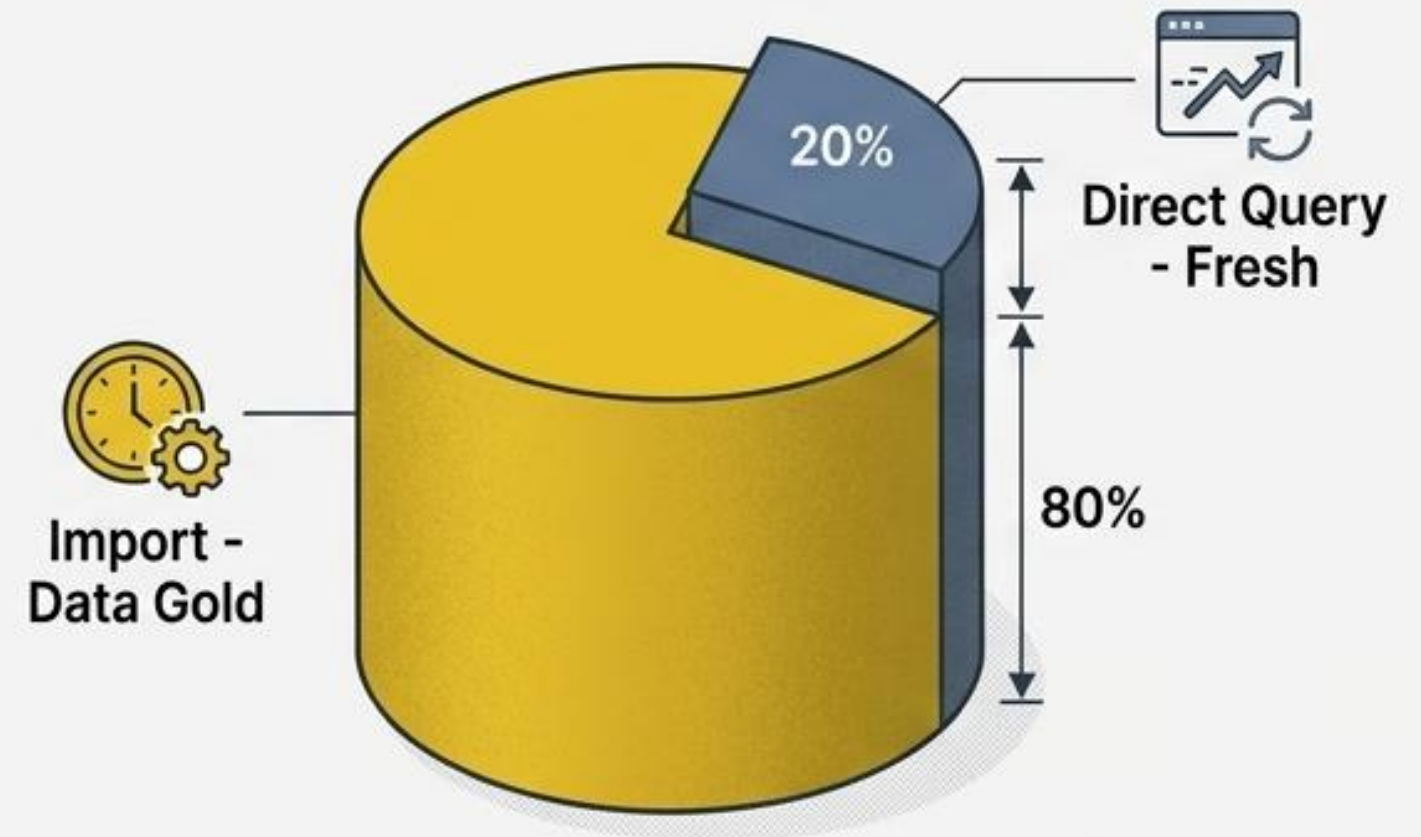
Advanced Storage Architecture

Live Connection



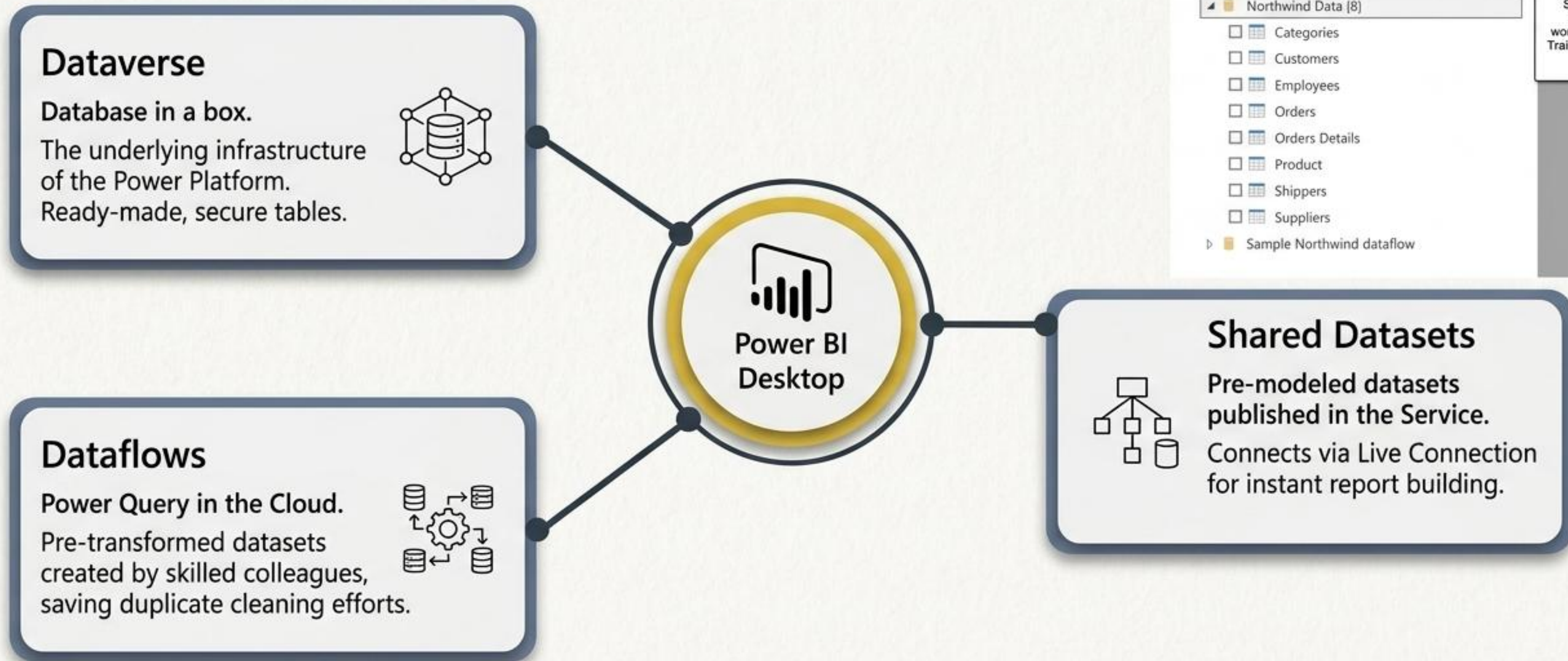
Connects directly to pre-modeled Analysis Services (Tabular/Multi) or Power BI Service Datasets.
No local modeling required.

Hybrid Tables Model



Partitions a single table to get the best of both worlds:
Speedy performance for historical data (Import) +
latest data up-to-the-minute (Direct Query).

Connecting to Enterprise Single-Sources-of-Truth



A **dataflow** appears the same way a worksheet, a table, or another data source appears. Note that you do have to access the location of the **dataflow**, usually a workspace in the Service. In this example, the workspace name is Training Workspace.

Bulk Extraction: The Folder Funnel

Metadata Columns

Name

Extension

Date accessed

Date modified

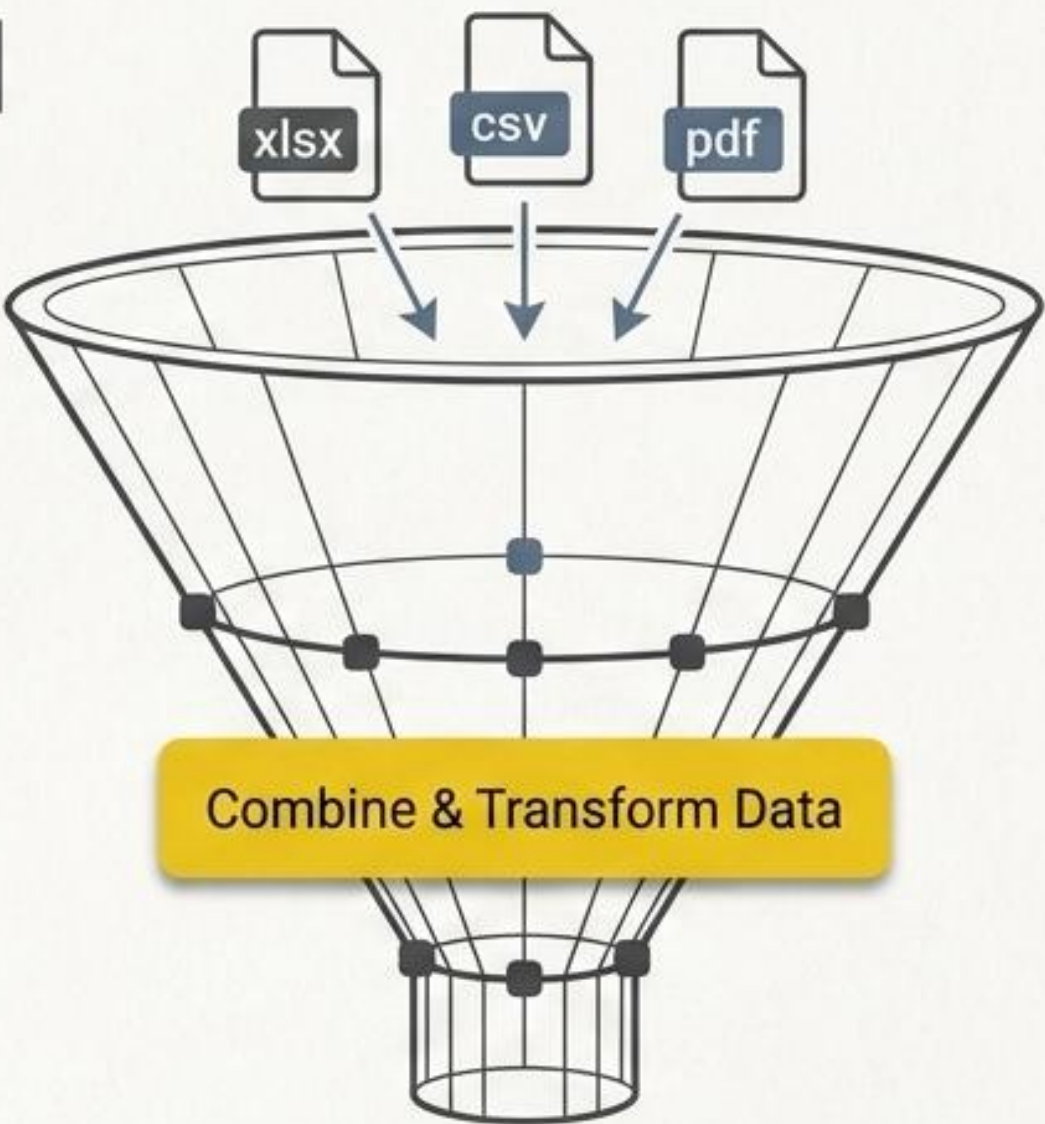
Date created

Combine Buttons

Combine & Transform Data

Transform Data

Extract Metadata:
Connect to a Folder to pull
a list of file properties
(Name, Extension, Date)



Name	Extension	Date	Date
1	xlsx	5/15/2022	3/16/2020
2	csv	5/15/2022	3/14/2020
3	csv	5/15/2023	5/15/2020
4	pdf	5/19/2023	5/8/2020

Expands the hidden
binaries into a single,
unified query.



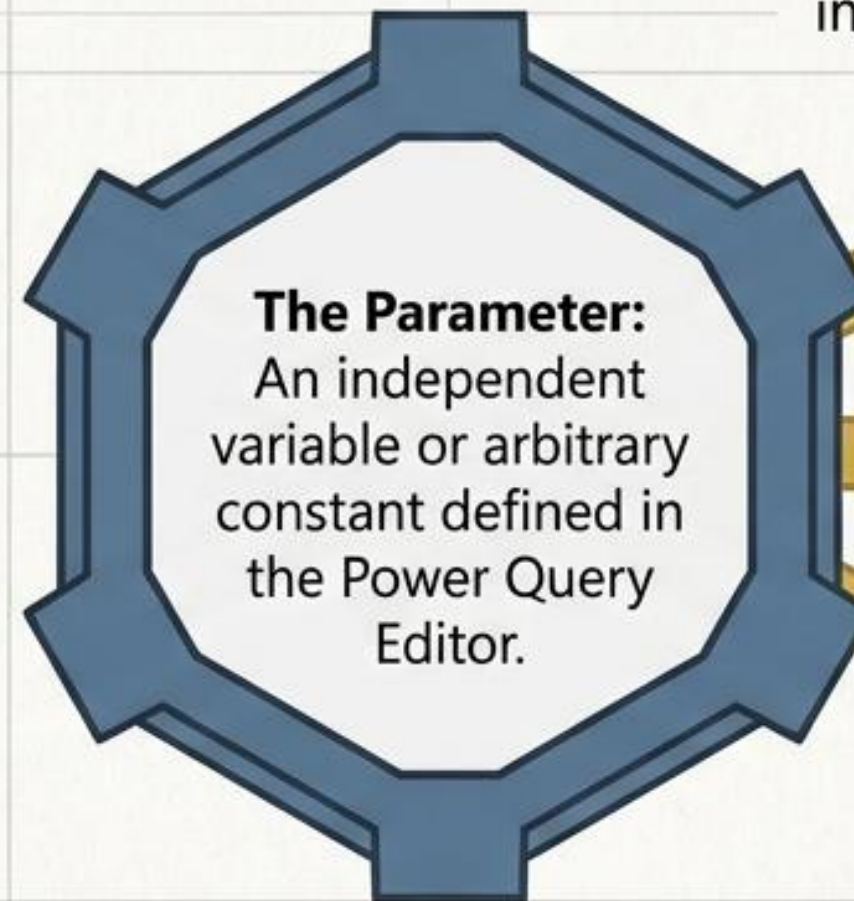
Critical Constraint: Files must have the exact same structure (column names, column count) or the query will require complex, messy transformations.

Demystifying Parameters

Three Core Functions

- 1** Provides a dynamic value to a function within Power Query.
- 2** Allows a user to safely alter values in Power BI Desktop.
- 3** Allows users to update values in the Power BI Service.

The Parameter Injection Engine



Define once,
inject everywhere



Building the Parameter Engine

The image shows two screenshots of a 'Manage Parameters' dialog box. The left screenshot shows the 'New' button highlighted in the top left corner. The right screenshot shows the 'SampleParameter' entry selected in the list on the left, and the 'Suggested Values' dropdown menu is open, showing 'Query' and 'Country-Dp' as options. The 'Current Value' field contains the text 'Select a Country-Zip combination'.

Step 1: Name & Identity

Navigate to Manage Parameters -> New. Provide a unique name and clear description.

Step 2: Data Type

Lock in the structural data type (e.g., Text for paths, Date for times).

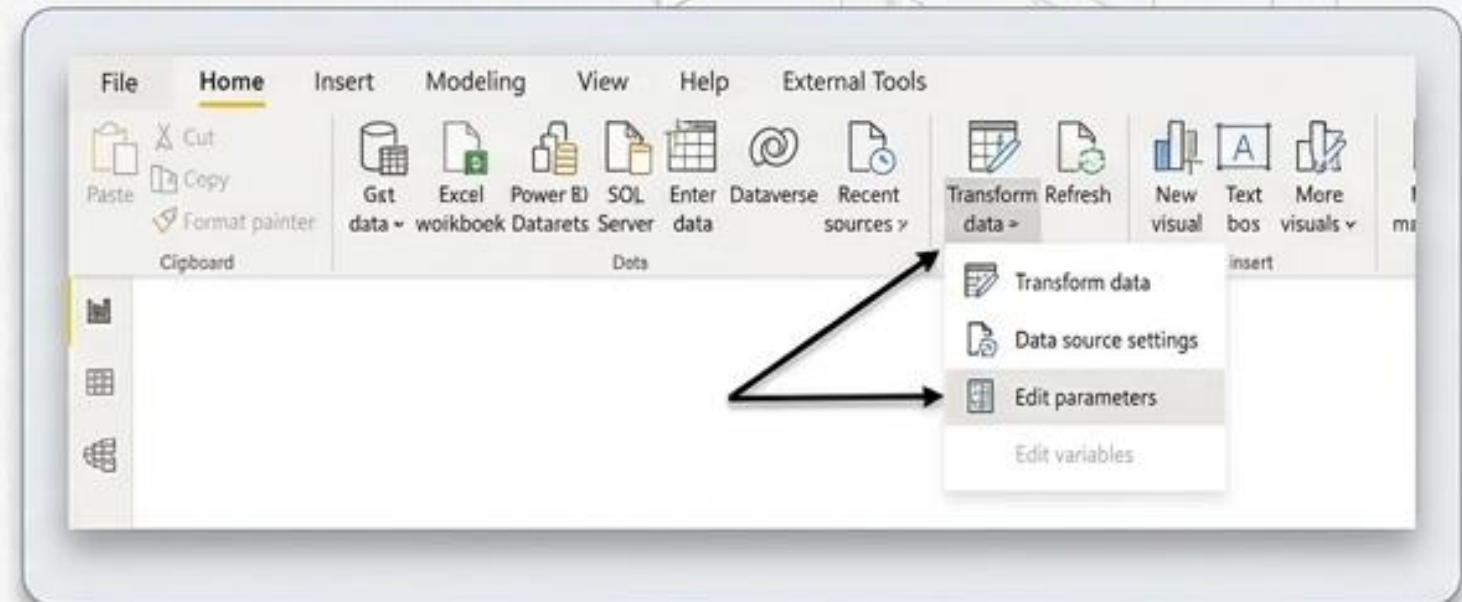
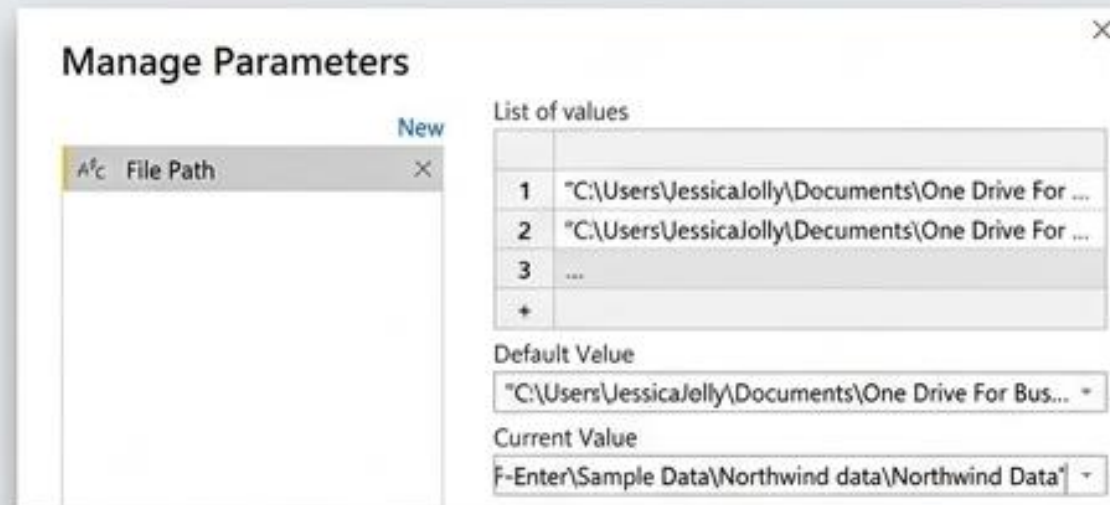
Step 3: Suggested Values

Choose 'Any Value', a custom 'List of Values', or link an existing 'Query'.

Step 4: Default/Current Value

Set the baseline value or provide a helpful prompt message for the end user.

Real-World Parameter Implementation



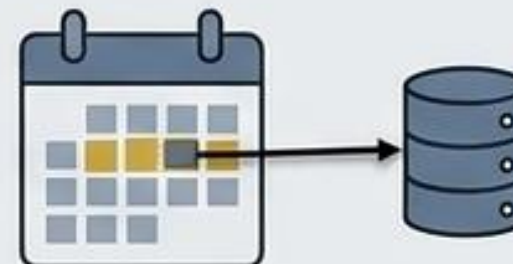
Dynamic File Paths

Create a list parameter of file/server paths.



Crucial: Right-click the parameter and select 'Enable Load' so it appears in PBI Desktop under Transform Data -> Edit Parameters.

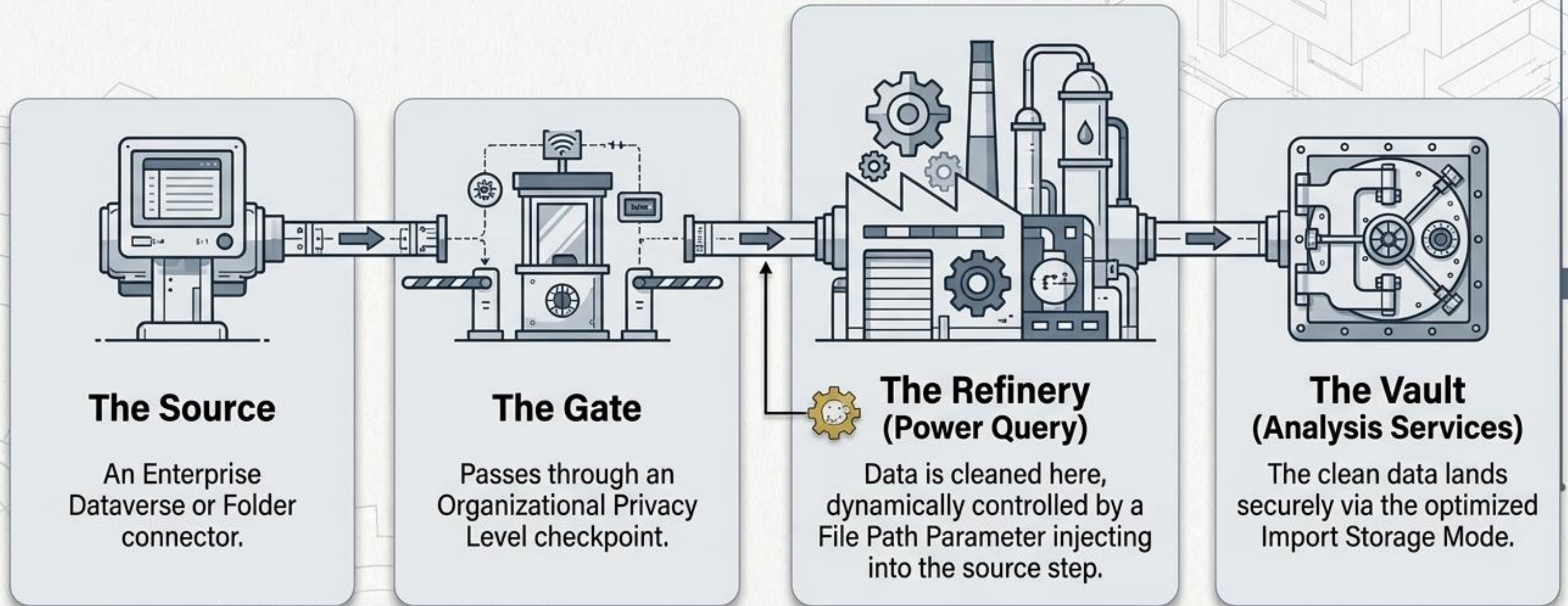
Incremental Refresh



Create RangeStart and RangeEnd Date/Time parameters.

This is a strict prerequisite for configuring Incremental Refresh in the Service, ensuring you only load new/changed data rather than the whole history.

Synthesis: The Anatomy of a Resilient Pipeline



The Perfect Query: The right connector, transformed immediately, securely gated, dynamically parameterized, and optimally stored.